



SACC

2020 中国系统架构师大会

SYSTEM ARCHITECT CONFERENCE CHINA 2020

架构融合 云化共建

LIVE 2020年10月22日 - 24日网络直播

架构融合
云化共建

eBay在**亿万级**广告大数据**场景**下基于DDDD的**响应**式数据管道**实践**

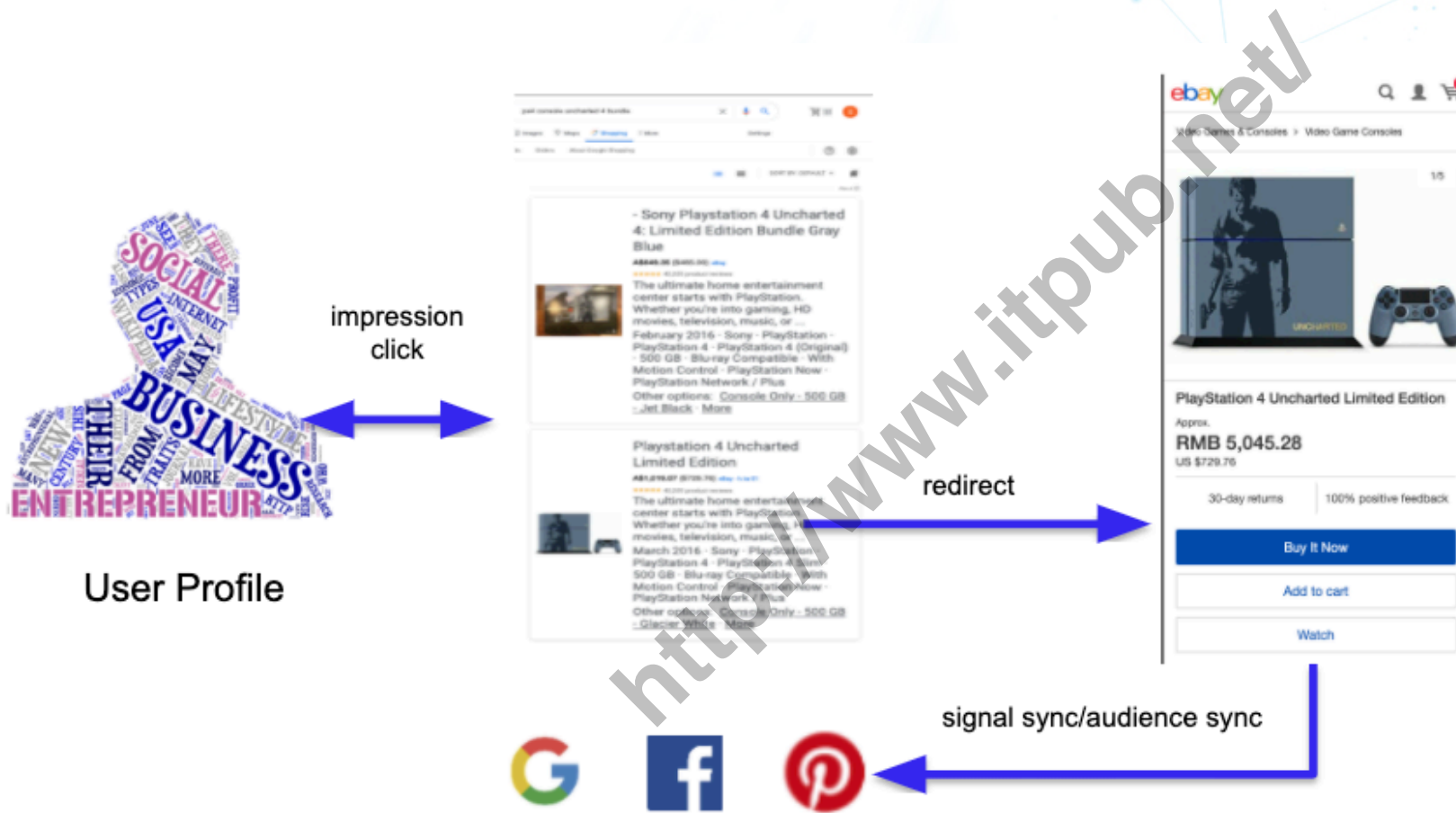
张申竣 - eBay MTS2 Engineer

Business Overview

Create & deliver audience lists, **audience targeting**, provide **user insights** and facilitate near real time **user level sync** across all partners.

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Business Value



Metrics

Profitable Business

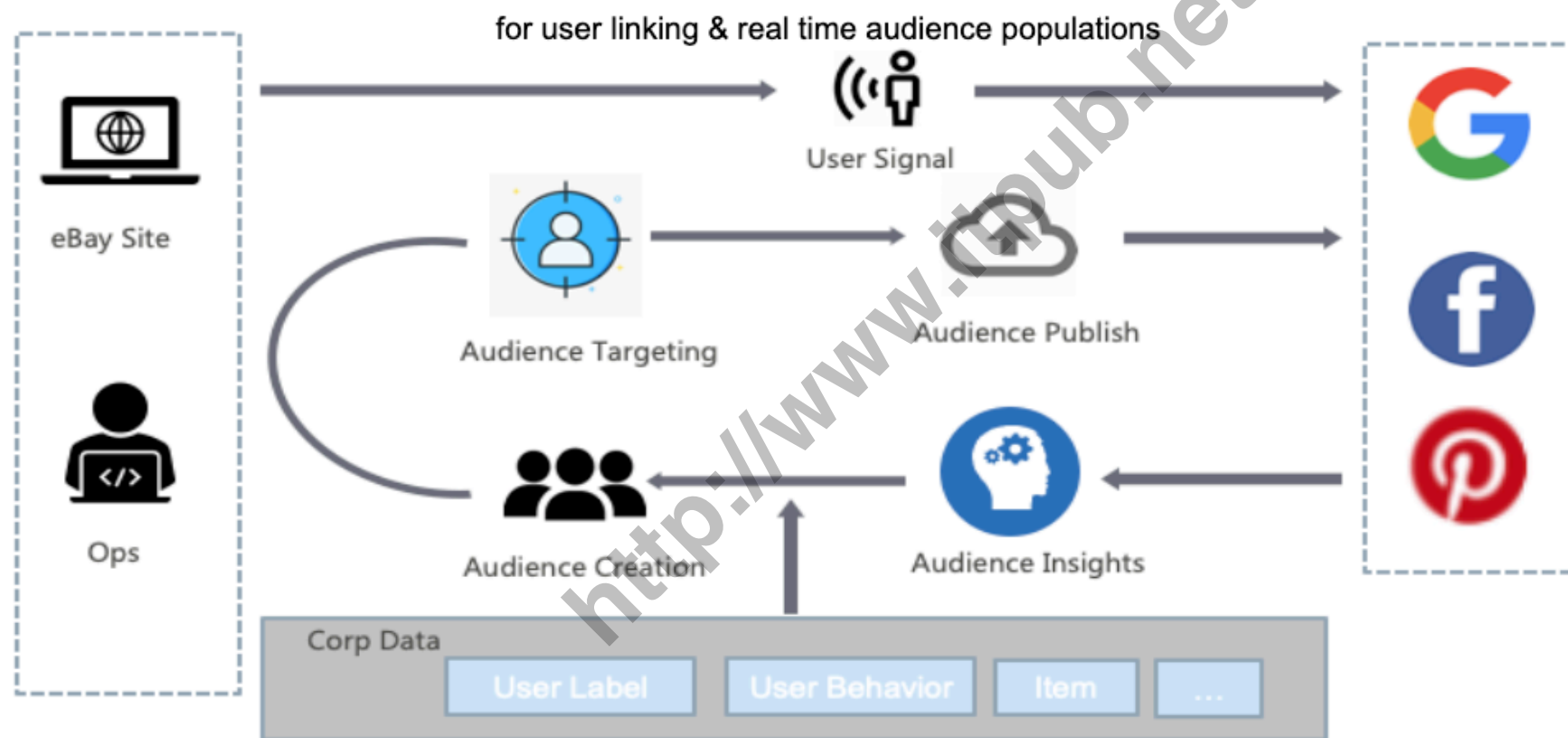
- Hundreds of millions \$ business
- Double digit ROI
- Growing (double digit growth)

Scaled Big Data System

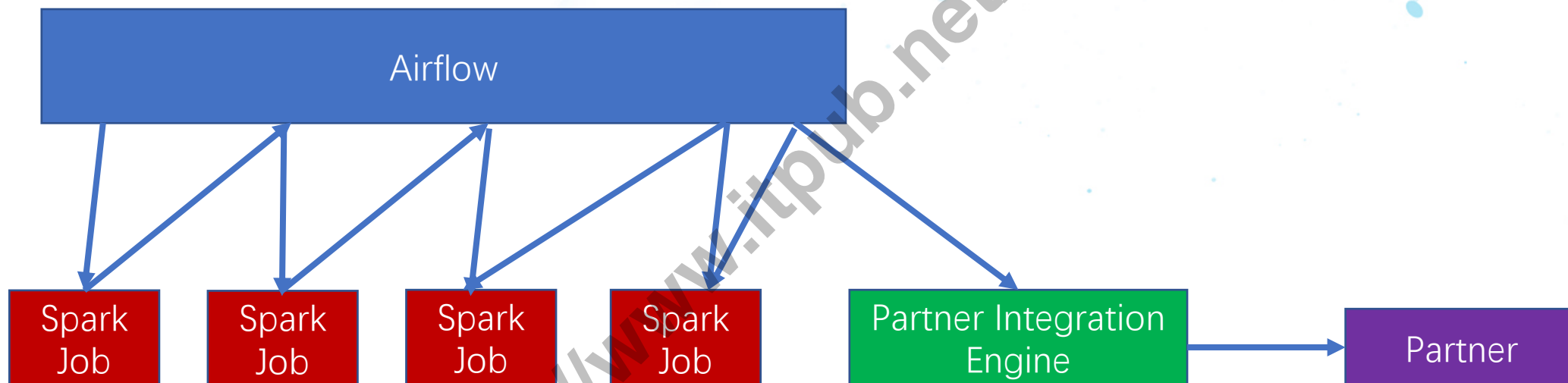
- TB Level Daily Data Processing
- ~10K/s QPS for NRT Processing

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Engineer Method



Architecture 1.0



Pain Point

1.0's Implementation

1. Airflow to schedule the batch job after new audience user list ready in Audience Platform and run the shell for each step of an RLSA pipeline
2. Shell to handle the core business logic
3. Spark job is submitted by shell to handle the rest of the business logic and user data enrichment/filtering
4. Partner Integration Engine will be triggered by shell to upload the user list

Pain Point

1. Lack of domain knowledge, it's impossible to support new audience types and more partners without the domain knowledge
2. Core business logic is written in shell, it's impossible to extend.
3. Spark job handles both metadata (business logic) calculation and big data calculation.
4. In most cases, partner audience operation is an asynchronized reactive system involve many other external system like audience platform, campaign manager and partner integration engine. But the current implementation is more like a synchronized system depends on the status of a shell script. It's not easy to locate the issue for each step and restart a single step for the whole pipeline.

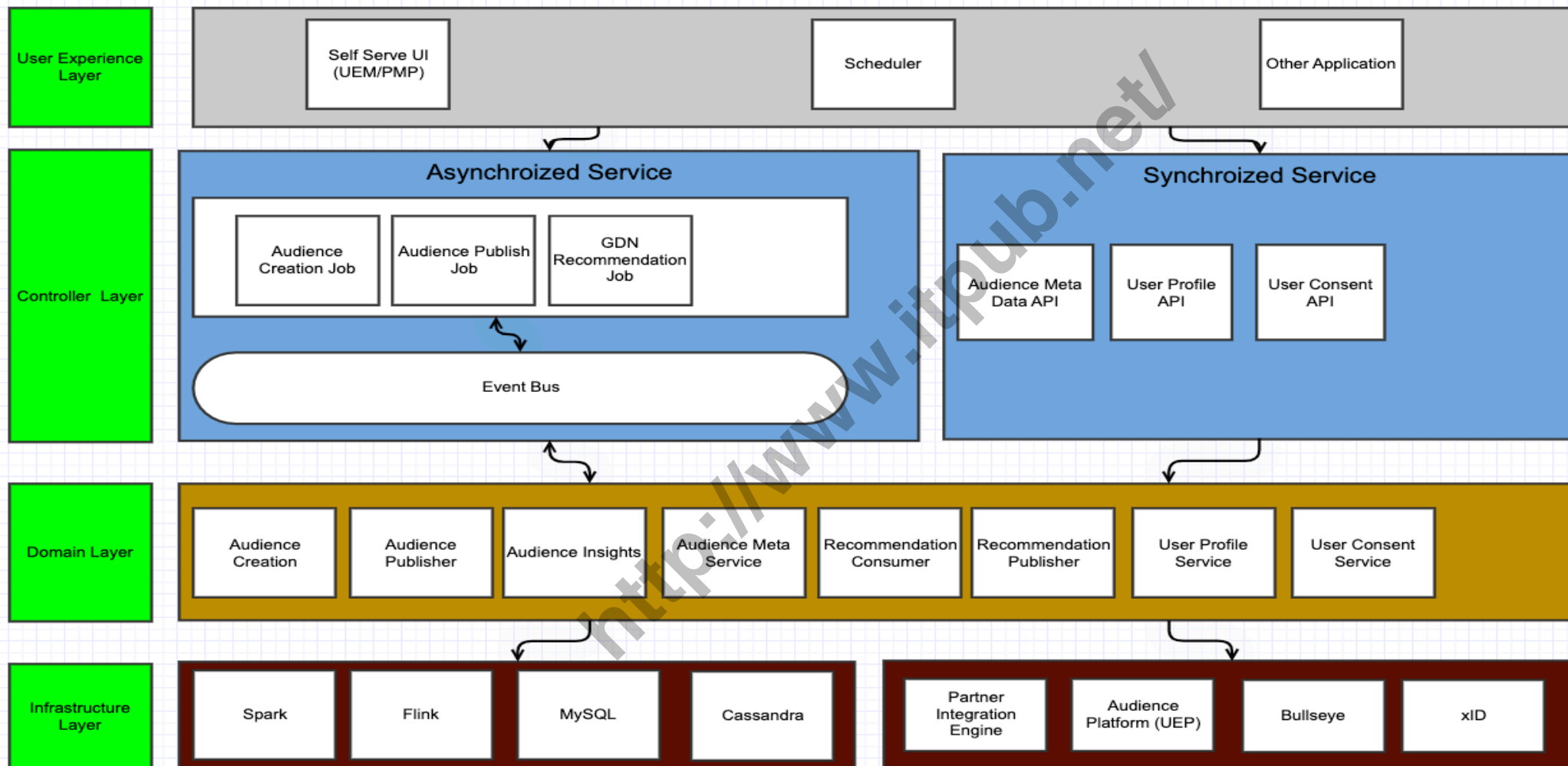
Evolvment

Architecture 2.0

1. Leverage Domain Driven Development (DDD) to bring the domain knowledge to support more audience types and channels
2. Use event sourcing pattern to build an asynchronized reactive system to improve the user experience and easy to debug.
3. The core domain layer is built on Spring Boot and spark job is focused on the big data processing.

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Architecture 2.0



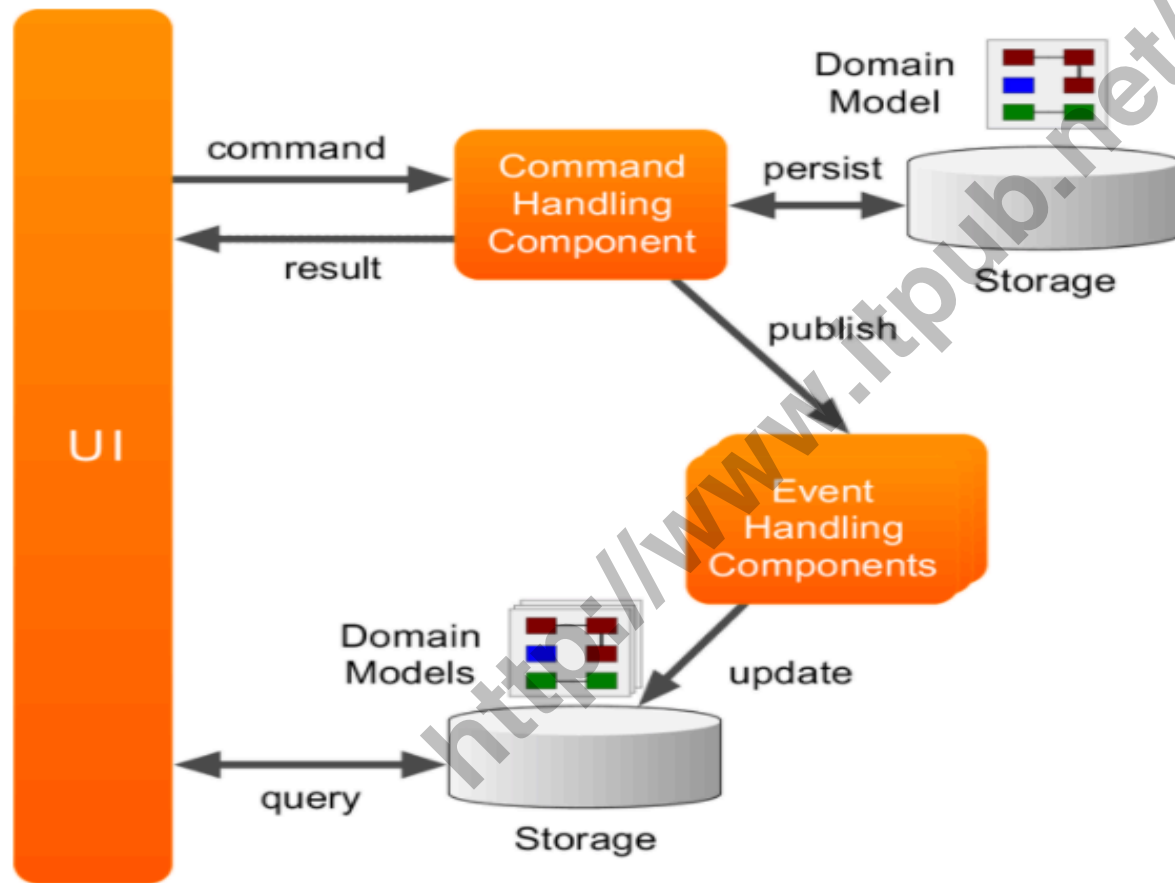
Tech Stack



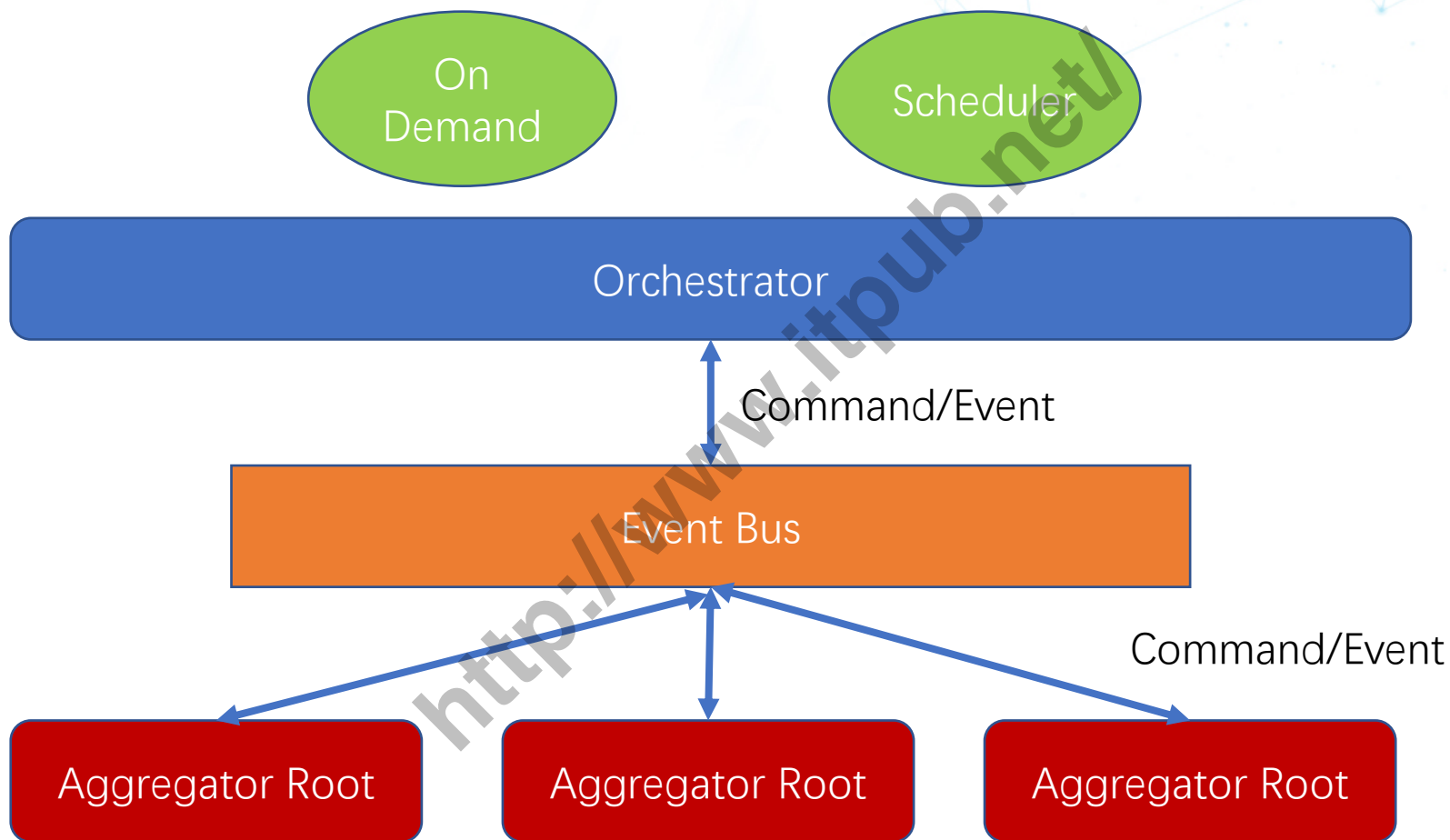
Open Source Framework for DDD

Name	Language Support	Document	Feature	License
Axon	Java	Comprehensive, Update Frequently	Comprehensive, CQRS/ES/Saga, Easy to Integrate with Spring and Distributed Message Queue, Enterprise Level	Axon Framework: Apache 2 Axon Server (Standard): AxonIO-Specific Open Source License Axon Server (Enterprise): Commercial
Eventuate	Java/Scala	Rough	Lightweight, CQRS/ES/Saga	Apache 2
Akka Persistence	Scala/.Net	Comprehensive, Update Frequently	Lightweight, Akka Actor Based, CQRS/ES, In JVM Message Bus	Apache 2

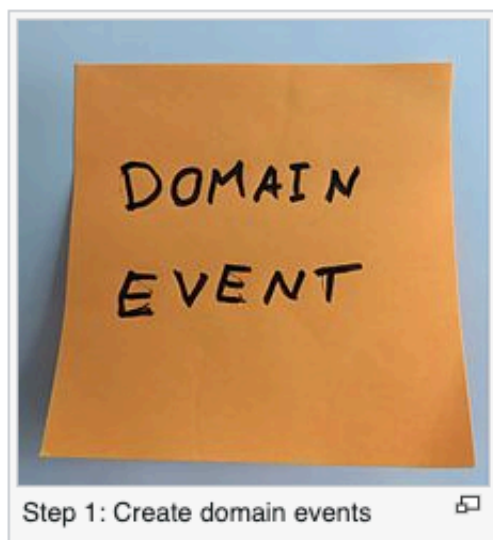
Traditional CQRS in DDD



Our Use Case



Event Storming

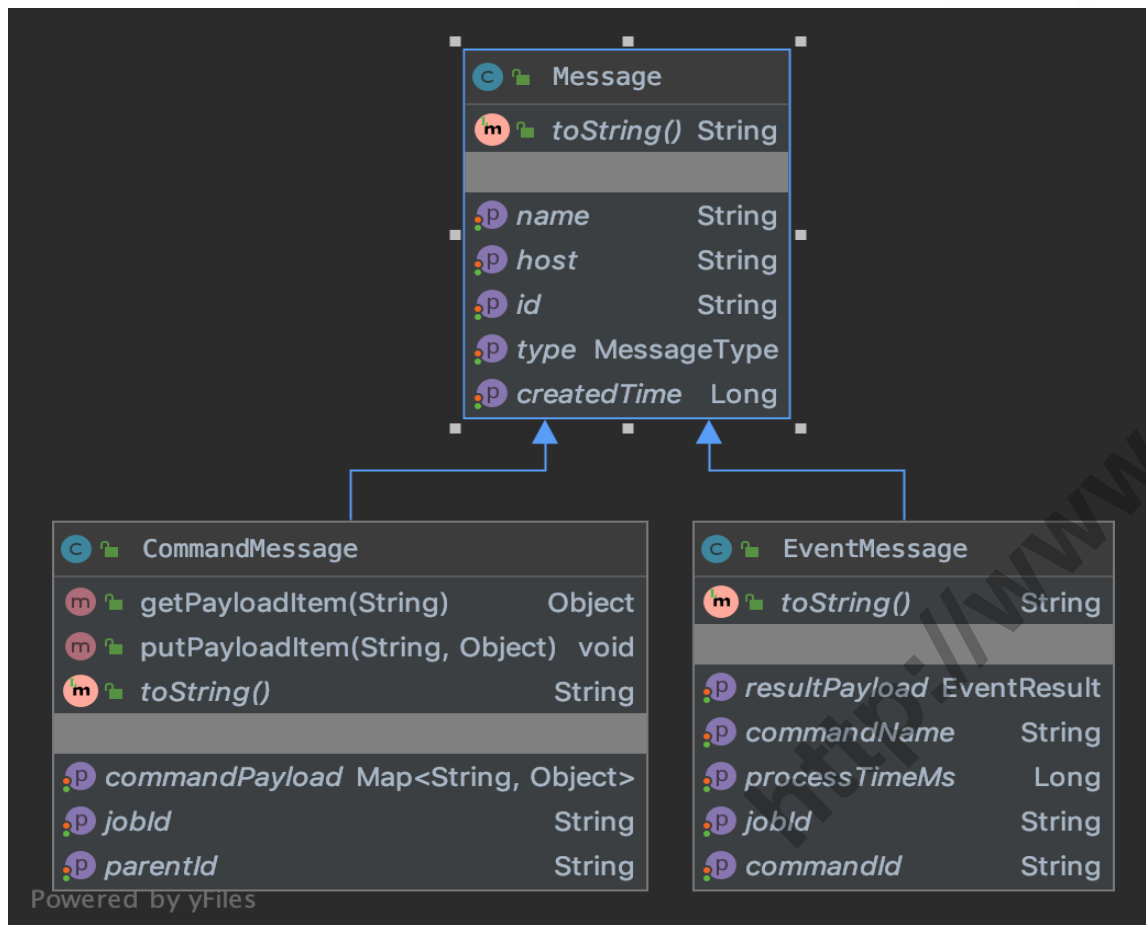


DDD: What We Need

- Unified Message Schema
- Extensible Command & Event
- A Transparent Way to Handle Command
- A Transparent Way to Publish the Event
- Easy Programming Model

A Lightweight Framework to Help Developer to Use
Event Storming Methodology.

Unified Message Schema



Annotation Based Aggregator Root

```
@CommandHandler(name = "validation_tcommand")
@ResponseEvent(name = "tcommande", routingKey = "eventsourcing.event", routingSuffix = "index")
public EventResult createAudience(@CommandParam(name = "arg_1") int arg1,
@CommandParam(name = "arg_2") String arg2,
    @Valid TestValidationCommand testValidationCommand, BindingResult
    bindingResult
){
    BasicEventResult result = new BasicEventResult();
    if (bindingResult.hasErrors()) {
        result.setMessage("fail");
        result.setStatus(BasicCommandResultStatus.INVALID_COMMAND_PARAMETER);
    } else {
        //TODO Business Logic
        result.setMessage("success");
        result.setStatus(BasicCommandResultStatus.SUCCESS);
    }
    return result;
}
```

Self R&D Event Sourcing Framework

- Annotation Based Aggregator, Command Handler and Event Publisher
- Annotation Based Command Parameter Extractor
- Annotation Based Event Routing Key
- JSR-380 Support
- Pluggable Event Publisher
- Extensible for Different Message Broker

Demo

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Future Planning

- Self R&D Orchestrator Based on Event Sourcing Framework
- Saga
- Command Idempotent
- In Memory Message Bus
- Event Sourcing UI
- ...

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Q&A

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Thanks

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